Appendix: Fundamentals of Hamiltonian Monte Carlo for Bayesian Posterior Sampling

1 From Markov Chains to Metropolis–Hastings

We wish to draw from a target $\pi(x) = f(x)/Z$ on $\mathbb{R}^D$, where f can be evaluated pointwise but the normaliser $Z = \int f(x)\,dx$ cannot. MCMC gives up on sampling π directly and instead builds a Markov chain that visits states in proportion to π.

Definition (Markov chain and invariance)**.** A Markov chain on $\mathbb{R}^D$ is specified by a transition kernel $T(x \to x')$, the density of moving to x' from x. A distribution π is *invariant* (stationary) for T if $\int \pi(x)\,T(x \to x')\,dx = \pi(x')$. The design problem is to construct a T whose invariant distribution is the π we want.

Metropolis–Hastings. From the current state x, draw a proposal $x^* \sim q(x^* \mid x)$ and accept it with probability

$$\alpha(x \to x^*) = \min\left(1,\ \frac{f(x^*)\,q(x \mid x^*)}{f(x)\,q(x^* \mid x)}\right),$$

otherwise remain at x. The target enters only through the *ratio* $f(x^*)/f(x)$, so the intractable Z cancels. This is the single fact that makes Bayesian computation possible at all.

Definition (Detailed balance)**.** T satisfies detailed balance with respect to π if $\pi(x)\,T(x \to x') = \pi(x')\,T(x' \to x)$ for all x, x'.

Proposition. Detailed balance implies invariance: integrating both sides over x gives $\int \pi(x)T(x \to x')\,dx = \pi(x')\int T(x' \to x)\,dx = \pi(x')$, since $T(x' \to \cdot)$ is a density. Metropolis–Hastings is constructed to satisfy detailed balance, which is why it is correct for *any* proposal q.

The curse of dimensionality. Correct is not efficient. In high dimension the posterior mass does not sit near the mode but concentrates on a thin shell — the *typical set* — at radius $O(\sqrt{D})$, since the density falls off while the volume element grows like r^{D-1}. An isotropic random-walk proposal $\mathcal{N}(x, \epsilon^2 I)$ almost always steps off that shell into negligible density, so acceptance decays exponentially in D unless $\epsilon \propto D^{-1/2}$. Shrinking ϵ makes the chain diffuse: the iterations needed per effectively independent draw grow like $O(D)$. The sampler can only shuffle along a shell it should be traversing.

2 Hamiltonian Monte Carlo

HMC removes the random walk by giving the chain *momentum*, so that a single proposal can traverse the typical set rather than jitter within it.

Phase space. Augment the position $\boldsymbol{\theta} \in \mathbb{R}^D$ with an auxiliary momentum $\mathbf{p} \sim \mathcal{N}(\mathbf{0}, \mathbf{M})$, giving a $2D$-dimensional phase space. Define the *Hamiltonian*

$$H(\boldsymbol{\theta}, \mathbf{p}) = U(\boldsymbol{\theta}) + K(\mathbf{p}), \qquad U(\boldsymbol{\theta}) = -\ln f(\boldsymbol{\theta}), \qquad K(\mathbf{p}) = \tfrac{1}{2}\mathbf{p}^\top \mathbf{M}^{-1}\mathbf{p}.$$

The joint density $\pi(\boldsymbol{\theta}, \mathbf{p}) \propto \exp\{-H(\boldsymbol{\theta}, \mathbf{p})\}$ factorises, so its $\boldsymbol{\theta}$-marginal is exactly the target. Momentum is scaffolding: it is resampled each iteration and discarded.

Intuition. U is a landscape whose valleys are regions of high posterior density, and the state is a frictionless puck on it. Kicked in a random direction, the puck climbs slopes and turns back under its own momentum, sweeping along a contour of constant total energy — covering long distances *within* the typical set instead of jittering across it.

Definition (Hamilton's equations)**.**

$$\frac{d\boldsymbol{\theta}}{dt} = \frac{\partial H}{\partial \mathbf{p}} = \mathbf{M}^{-1}\mathbf{p}, \qquad \frac{d\mathbf{p}}{dt} = -\frac{\partial H}{\partial \boldsymbol{\theta}} = -\nabla U(\boldsymbol{\theta}).$$

Write Φ_t for the flow map carrying $(\boldsymbol{\theta}(0), \mathbf{p}(0))$ to $(\boldsymbol{\theta}(t), \mathbf{p}(t))$.

Theorem (Conservation of energy)**.** Along any solution, $\frac{dH}{dt} = 0$; hence $H(\boldsymbol{\theta}(t), \mathbf{p}(t)) = H(\boldsymbol{\theta}(0), \mathbf{p}(0))$ for all t.

Theorem (Liouville's theorem)**.** The vector field of Hamilton's equations is divergence-free, so Φ_t preserves phase-space volume: $\det J_{\Phi_t} = 1$ for all t.

These two facts are what a Metropolis correction needs. A proposal generated by running Φ_L then negating the momentum is a deterministic involution, hence reversible, with acceptance ratio

$$\alpha = \min\Big(1,\ \exp\{-H(\boldsymbol{\theta}^*, \mathbf{p}^*) + H(\boldsymbol{\theta}_0, \mathbf{p}_0)\} \cdot \big|\det J_{\Phi_L}\big|\Big).$$

Liouville's theorem sets the Jacobian to 1; conservation of energy sets the exponent to 0. So an *exact* Hamiltonian trajectory is accepted with probability 1, no matter how far it travels.

3 Bayesian Inference and the Complete Algorithm

The posterior challenge. For parameters $\boldsymbol{\theta}$ and data $\mathcal{D}$,

$$p(\boldsymbol{\theta} \mid \mathcal{D}) = \frac{p(\mathcal{D} \mid \boldsymbol{\theta})\, p(\boldsymbol{\theta})}{\int p(\mathcal{D} \mid \boldsymbol{\theta})\, p(\boldsymbol{\theta})\, d\boldsymbol{\theta}} = \frac{f(\boldsymbol{\theta})}{Z}.$$

The evidence $Z = p(\mathcal{D})$ is a D-dimensional integral: grid quadrature costs $O(k^D)$, and conjugate updates do not exist for hierarchical models with non-conjugate likelihoods. But HMC needs only

$$U(\boldsymbol{\theta}) = -\ln p(\mathcal{D} \mid \boldsymbol{\theta}) - \ln p(\boldsymbol{\theta}),$$

and its gradient, both computable without Z.

Leapfrog integration. Hamilton's equations rarely have closed-form solutions, so the flow is approximated in steps of size ϵ by the *leapfrog* (Störmer–Verlet) scheme — half a momentum kick, a full position drift, and a second half kick:

$$\begin{aligned}
\mathbf{p}(t + \tfrac{\epsilon}{2}) &= \mathbf{p}(t) - \tfrac{\epsilon}{2}\nabla U(\boldsymbol{\theta}(t)),\\
\boldsymbol{\theta}(t + \epsilon) &= \boldsymbol{\theta}(t) + \epsilon\, \mathbf{M}^{-1}\mathbf{p}(t + \tfrac{\epsilon}{2}),\\
\mathbf{p}(t + \epsilon) &= \mathbf{p}(t + \tfrac{\epsilon}{2}) - \tfrac{\epsilon}{2}\nabla U(\boldsymbol{\theta}(t + \epsilon)).
\end{aligned}$$

Theorem (Symplecticity and the shadow Hamiltonian)**.** Each leapfrog step is a composition of shear maps, so it preserves phase volume *exactly* ($\det J = 1$) and is time-reversible. Being symplectic, it conserves exactly a nearby *shadow Hamiltonian* $\tilde{H} = H + \mathcal{O}(\epsilon^2)$, so energy error stays bounded over arbitrarily long trajectories instead of accumulating. Forward Euler, by contrast, has $\det J \neq 1$ and no shadow Hamiltonian: its energy drifts monotonically and trajectories spiral away.

Because volume preservation and reversibility hold *exactly* for leapfrog, the Metropolis correction below restores exactness for any ϵ; the step size affects only efficiency, through the acceptance rate.

Hamiltonian Monte Carlo
Input: initial state $\boldsymbol{\theta}^{(0)}$, trajectory length L, step size ϵ, mass matrix $\mathbf{M}$.
For $s = 1, \ldots, S$:
1. Sample fresh momentum $\mathbf{p}^{(s)} \sim \mathcal{N}(\mathbf{0}, \mathbf{M})$.
2. Set $(\boldsymbol{\theta}_0, \mathbf{p}_0) = (\boldsymbol{\theta}^{(s-1)}, \mathbf{p}^{(s)})$.
3. Run L leapfrog steps of size ϵ using $\nabla U(\boldsymbol{\theta})$ to obtain $(\boldsymbol{\theta}^*, \mathbf{p}^*)$.
4. Compute $\alpha = \min\big(1, \exp\{-H(\boldsymbol{\theta}^*, \mathbf{p}^*) + H(\boldsymbol{\theta}_0, \mathbf{p}_0)\}\big)$.
5. With probability α set $\boldsymbol{\theta}^{(s)} = \boldsymbol{\theta}^*$; otherwise $\boldsymbol{\theta}^{(s)} = \boldsymbol{\theta}^{(s-1)}$.

Output: draws $\boldsymbol{\theta}^{(1)}, \ldots, \boldsymbol{\theta}^{(S)}$.

From draws to summaries. The output is an $S \times D$ matrix whose rows are dependent but marginally distributed as the posterior.

Theorem (Ergodic theorem for MCMC)**.** If the chain is irreducible and aperiodic with invariant distribution $p(\boldsymbol{\theta} \mid \mathcal{D})$, then for any integrable g,

$$\frac{1}{S}\sum_{s=1}^{S} g\big(\boldsymbol{\theta}^{(s)}\big) \xrightarrow[S\to\infty]{\text{a.s.}} \mathbb{E}_{p(\boldsymbol{\theta}\mid\mathcal{D})}\big[g(\boldsymbol{\theta})\big].$$

This licenses every number in the main report. For the multilevel model $\alpha_j \sim \mathrm{Normal}(\bar{\alpha}, \sigma)$, a listing's posterior mean intercept is the column average of that α_j, an 89% credible interval is the corresponding pair of column quantiles, and a derived quantity — $\mathrm{logit}^{-1}(\alpha_j)$, or a prediction for an unseen listing $\mathrm{logit}^{-1}(\bar{\alpha} + z\sigma)$ — is estimated by transforming row-wise and *then* averaging. The order matters: averaging first estimates a different quantity, since logit^{-1} is non-linear.

Two caveats follow. The guarantee is asymptotic, so n_{eff} must be reported alongside it. And a *divergent transition* signals that leapfrog has broken down: where the posterior curves more sharply than ϵ can resolve, $\tilde{H}$ stops tracking H and the region is under-explored, so divergences indicate bias rather than mere inefficiency — which is why the main report reparameterises instead of ignoring them.